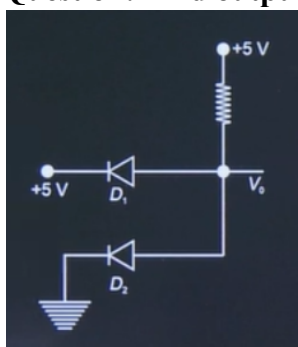


**JEE-Main-08-04-2025 (Memory Based)**  
**[EVENING SHIFT]**  
**Physics**

**Question:** Find output voltage in the given circuit.



**Options:**

- (a) +5 volt
- (b) Zero
- (c) 10 volt
- (d) -5 volt

**Answer: (b)**

**Question:** A fractional errors in  $x$ ,  $y$  and  $z$  are 0.1, 0.2 and 0.5 respectively. Find

maximum fractional error in  $x^{-2}y^2z^{\frac{3}{5}}$ .

**Options:**

- (a) 0.2
- (b) 0.7
- (c) 0.6
- (d) 0.3

**Answer: (b)**

**Question:** For a nucleus of mass number  $A$  and radius  $R$ , mass density  $\rho$ . Then choose the correct option

**Options:**

- (a)  $\rho \propto A^{1/3}$
- (b)  $\rho$  is independent of  $A$
- (c)  $\rho \propto A$
- (d)  $\rho \propto A^3$

**Answer (b)**

**Question:** A convex lens ( $f = 30$  cm) is in contact with concave lens ( $f = 20$  cm). Object is placed on the left side at a distance of 20 cm. Find the image distance.

**Options:**

- (a) 10 cm
- (b) 20 cm
- (c) 15 cm
- (d) 25 cm

**Answer:** (c)

**Question:** There are two charged sphere of radius  $R$  and  $3R$ . When the sphere are made to touch each other and then separate, the surface charge density becomes  $r_1$  and  $r_2$

respectively. Find  $\frac{r_1}{r_2}$ .

**Options:**

- (a)  $1/9$
- (b)  $1/3$
- (c) 3
- (d) 9

**Answer:** (c)

**Question:** Given  $\lambda = \frac{2nC}{m}$  (linear charge density) is for  $Q$  wire which is passing through body diagonal of a closed cube of side length  $\sqrt{3}$  cm. Find flux through the cube.

**Options:**

- (a)  $1.44\pi$
- (b)  $0.72\pi$
- (c)  $2.16\pi$
- (d)  $6.84\pi$

**Answer:** (c)

**Question:** A monatomic gas is stored in a thermally insulated container. The gas is

suddenly compressed to  $\left(\frac{1}{8}\right)^{\text{th}}$  of its initial volume. Find ratio of final pressure to initial pressure.

**Options:**

- (a) 8
- (b) 16
- (c) 4
- (d) 32

**Answer:** (d)

**Question:** Two balls are projected with same speed at different angles. If maximum height of 1st is 8 times maximum height of 2nd ball. Find the ratio of their time of flight.

**Options:**

- (a)  $1 : 2\sqrt{2}$
- (b)  $2\sqrt{2} : 1$

- (c) 2 : 1  
(d) 4 : 1

**Answer: (b)**

**Question:** A uniform disc of radius  $r$  is rotating about an axis passing through diameter with angular speed 800 rpm. A torque of magnitude  $25\pi$  Nm is applied on the disc for 40 sec. If final angular speed of disc is 2100 rpm. Find radius of the disc if mass is 1 kg.

**Options:**

- (a)  $40/3$   
(b)  $20\sqrt{\frac{3}{13}}$   
(c)  $20\sqrt{\frac{2}{13}}$   
(d)  $10\sqrt{\frac{3}{2}}$

**Answer: (b)**

**Question:** Water falls from 200 m height. What is increase in temperature when it touches the bottom. (Assume that all the heat goes into same amount of mass which was falling).

**Options:**

- (a)  $0.7^\circ\text{C}$   
(b)  $\frac{10}{21}^\circ\text{C}$   
(c)  $\frac{20}{21}^\circ\text{C}$   
(d)  $\frac{11}{10}^\circ\text{C}$

**Answer: (b)**

**Question:** Bulk modulus of a liquid is  $2 \times 10^9$  Pa initial and final pressure are 1 atm and 5 atm respectively. Find initial volume of the liquid if change in volume is  $0.8 \text{ cm}^3$ .

**Options:**

- (a)  $2 \times 10^3 \text{ cm}^3$   
(b)  $4 \times 10^3 \text{ cm}^3$   
(c)  $2 \times 10^{-4} \text{ cm}^3$   
(d)  $4 \times 10^{-3} \text{ cm}^3$

**Answer: (b)**

**Question:** The amplitude and phase of the wave when two travelling waves given as

$$y_1(x, t) = 4 \sin(\omega t - kx) \quad \text{and} \quad y_2(x, t) = 2 \sin\left(\omega t - kx = \frac{2\pi}{3}\right)$$

**Options:**

- (a)  $6, \frac{2\pi}{3}$
- (b)  $6, \frac{\pi}{3}$
- (c)  $2\sqrt{3}, \frac{\pi}{6}$
- (d)  $\sqrt{3}, \frac{\pi}{6}$

**Answer: (c)**

**Question:** An electron is released in the field generated by a non-conductivity sheet of uniform surface charge density  $\sigma$ . The rate of change of de-Broglie wavelength associated with electron waves inversely as nth power of distance travelled. Find the value of n

**Options:**

- (a)  $-1/2$
- (b)  $-1$
- (c)  $-1/4$
- (d)  $4$

**Answer: (b)**

**Question:** A force  $\hat{6k}$  is applied for  $\frac{5}{3}$  seconds on a body of mass 2 kg. If initial velocity of body was  $\hat{3i} + \hat{4j}$ . Then find final velocity of the body.

**Options:**

- (a)  $\hat{3i} + \hat{j} + \hat{5k}$
- (b)  $\hat{3i} + \hat{4j} + \hat{5k}$
- (c)  $\hat{3i} + \hat{2j} - \hat{3k}$
- (d)  $\hat{3i} + \hat{4j} - \hat{5k}$

**Answer: (a)**

**Question:** A rod of linear mass density ' $\lambda$ ' and length 'L' is bent into the form of a ring of radius R. Moment of inertia of ring about any of its diameter is

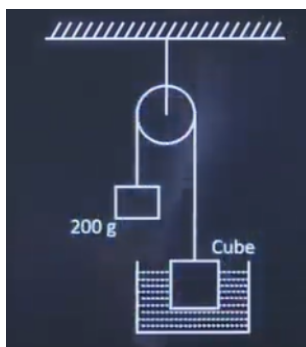
**Options:**

- (a)  $\frac{\lambda L^3}{12}$
- (b)  $\frac{\lambda L^3}{4\pi^2}$
- (c)  $\frac{\lambda L^2}{12}$

$$(d) \frac{\lambda L^3}{8\pi^2}$$

**Answer: (d)**

**Question:** A cube of side 10 cm is suspended from one end of a fine string of length 27 cm, and a mass of 200 gram is connected to the other end of the string. When the cube is half immersed in water then the system remains in balance. Find density of material of cube.



**Options:**

- (a) 800 kg/m<sup>3</sup>
- (b) 500 kg/m<sup>3</sup>
- (c) 700 kg/m<sup>3</sup>
- (d) 600 kg/m<sup>3</sup>

**Answer: (c)**

## JEE-Main-08-04-2025 (Memory Based)

## [EVENING SHIFT]

## Chemistry

**Question:** Consider the last electron of an element having atomic No. 9 & choose correct option

**Options:**

- (a) Sum of total nodes = 1
- (b)  $n = 2; l = 0$
- (c) Last electron enters in 2s subshell
- (d) There are  $5e^-$  with  $l = 0$

**Answer:** (a)

**Question:** Which of the following has  $sp^3d^2$  hybridisation?

**Options:**

- (a)  $[NiCl_4]^{2-}$
- (b)  $[Ni(CO)_4]$
- (c)  $SF_6$
- (d)  $[Ni(CN)_4]^{2-}$

**Answer:** (c)

**Question:** Atomic number of element with lowest first ionisation enthalpy is

**Options:**

- (a) 32
- (b) 19
- (c) 35
- (d) 87

**Answer:** (d)

**Question:** Consider the following statement

**Statement-I:**  $H_2Se$  is more acidic than  $H_2Fe$

**Statement-II:**  $H_2Se$  has higher bond dissociation Enthalpy

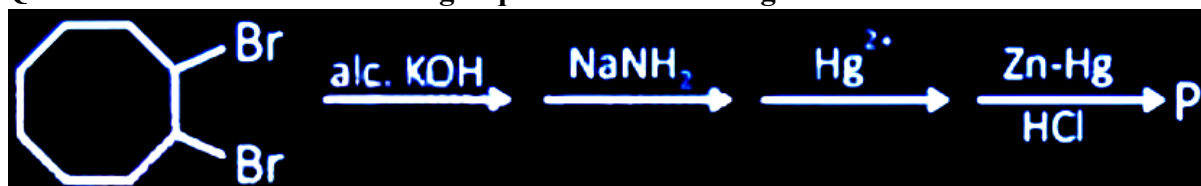
**In light of above statement, choose correct option**

**Options:**

- (a) Statement-I is true & Statement-II is false
- (b) Statement-I is false & Statement-II is true
- (c) Both Statement-I & Statement-II are true
- (d) Both Statement-I & Statement-II are false

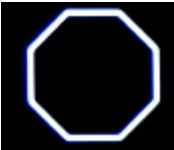
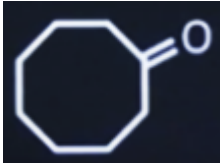
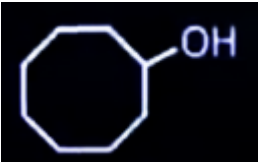
**Answer:** (a)

**Question:** Consider the following sequence of reactions given below



The product P is

Options:

- (a) 
- (b) 
- (c) 
- (d) 

Answer: (a)

Question: The correct IUPAC name of



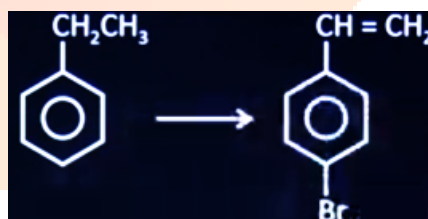
is

Options:

- (a) 4-ethylcyclopent-2-en-1-ol  
 (b) 3-ethylcyclopent-4-en-1-ol  
 (c) 5-ethylcyclopent-1-en-3-ol  
 (d) 3-ethylcyclopent-1-en-5-ol

Answer: (c)

Question: The correct sequence of reagent to be added for the following conversion



Options:

- (a)  $\text{Br}_2/\text{Fe}$ ; alc.  $\text{KOH}$ ;  $\text{Cl}_2/\text{FeCl}_3$   
 (b)  $\text{Br}_2/\text{FeCl}_3$ ;  $\text{Cl}_2/\Delta$ ; alc.  $\text{KOH}$   
 (c)  $\text{FeCl}_3/\text{Br}_2$ ; alc.  $\text{KOH}$ ;  $\text{H}^+/\Delta$   
 (d)  $\text{Cl}_2/\text{FeCl}_3$ ;  $\text{Br}_2/\text{FeCl}_3$ ; alc.  $\text{KOH}$

Answer: (b)

Question: For a first order reaction, the ratio of time required is  $t_1/t_2$ , if  $t_1$  is time consumed when reactant reaches  $1/4^{\text{th}}$  of initial concentration and  $t_2$  is the time when it reaches  $1/8^{\text{th}}$  of initial concentration

**Options:**

- (a)  $\frac{2}{3}$
- (b)  $\frac{3}{4}$
- (c)  $\frac{3}{2}$
- (d)  $\frac{4}{3}$

**Answer: (a)**



**Question:** The correct IUPAC name of the product is:-

**Options:**

- (a) 1-acetyl-2-methyl cyclohexene
- (b) 1-(2-methylcyclohexene-1-yl) ethanone
- (c) Cyclo-oct-2-en-1-one
- (d) 2-Cyclobutene-1-one

**Answer: (b)**

**Question: Match List-I with List-II and choose the correct option.**

| List-I           | List-II                             |
|------------------|-------------------------------------|
| (a) Nucleophile  | (i) Tetrahedral shape               |
| (b) Electrophile | (ii) Planar and $sp^2$ hybridized   |
| (c) Carbocation  | (iii) Species that accepts electron |
| (d) Carbanion    | (iv) Species that donate electron   |

**Options:**

- (a) a(i), b(ii), c(iv), d(iii)
- (b) a(iv), b(iii), c(ii), d(i)
- (c) a(iv), b(iii), c(i), d(ii)
- (d) a(iii), b(iv), c(ii), d(i)

**Answer: (b)**

**Question: Match List-I with List-II and choose the correct option.**

| List-I                                           | List-II               |
|--------------------------------------------------|-----------------------|
| (a) dil $\text{KMnO}_4$                          | (i) Unsaturation test |
| (b) $\text{FeCl}_3$ test                         | (ii) Alcoholic -OH    |
| (c) Liberate $\text{CO}_2$ with $\text{NaHCO}_3$ | (iii) Phenolic -OH    |
| (d) Ceric Ammonium nitrate test                  | (iv) Carboxylic Acid  |



**Options:**

- (a) A-I, B-IV, C-III, D-II
- (b) A-IV, B-I, C-III, D-II
- (c) A-I, B-III, C-IV, D-II
- (d) A-III, B-II, C-IV, D-I

**Answer: (c)**

**Question:** An aqueous solution of 0.1 M HA shows depression in freezing point of  $0.2^{\circ}\text{C}$ . If  $K_f(\text{H}_2\text{O}) = 1.86 \text{ K kg mol}^{-1}$  and assuming molarity = molality, find the dissociation constant of HA.

**Options:**

- (a)  $4.50 \times 10^{-5}$
- (b)  $6.25 \times 10^{-3}$
- (c)  $5.625 \times 10^{-4}$
- (d)  $2.65 \times 10^{-4}$

**Answer: (c)**

**Question:** Which of the following solutions can form minimum boiling azeotrope?

**Options:**

- (a)  $\text{C}_2\text{H}_5\text{OH} + \text{H}_2\text{O}$
- (b) n-heptane + n-hexane
- (c)  $\text{CH}_3\text{COOH} + \text{C}_5\text{H}_5\text{N}$
- (d)  $\text{C}_2\text{H}_5\text{Br} + \text{C}_2\text{H}_4\text{I}$

**Answer: (a)**

# JEE-Main-08-04-2025 (Memory Based)

## [EVENING SHIFT]

### Maths

**Question:** The number of rational terms in the binomial expansion of  $\left(5^{\frac{1}{2}} + 7^{\frac{1}{8}}\right)^{1016}$

**Options:**

- (a) 129
- (b) 128
- (c) 127
- (d) 130

**Answer: (b)**

$$\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots = \frac{\pi^4}{90}$$

$$\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots = \alpha$$

**Question:**  $\frac{1}{2^4} + \frac{1}{4^4} + \frac{1}{6^4} + \dots = \beta$  Then find  $\frac{\alpha}{\beta}$ .

**Options:**

- (a) 15
- (b) 14
- (c) 23
- (d) 18

**Answer: (a)**

**Question:** There are 12 points in a plane in which 5 are collinear such that no three of them are in straight line, then the number of triangles that can be formed from any 3 vertices from 12 points.

**Options:**

- (a) 220
- (b) 210
- (c) 230
- (d) 240

**Answer: (b)**

**Question:** Two lines  $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$  and  $\frac{x-\lambda}{3} = \frac{y-3}{4} = \frac{z-4}{5}$  has shortest distance  $\frac{1}{\sqrt{6}}$ .

If  $\lambda_1, \lambda_2$  are values of  $\lambda$ , then radius of circle passing through  $(0, 0), (\lambda_1, \lambda_2), (\lambda_2, \lambda_1)$  is \_\_\_\_.

**Answer: ()**

$$(1, 2, 3)(\lambda, 3, 4)$$

$$\frac{[\overrightarrow{AC}\overrightarrow{bd}]}{|\overrightarrow{b} \times \overrightarrow{d}|}$$

$$\begin{vmatrix} \lambda - 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$$

$$= (\lambda - 1)(-1) - 1(-2) + 1(-1)$$

$$\frac{|1 - \lambda + 2 - 1|}{\sqrt{1^2 + 2^2 + 1^2}} = \frac{1}{\sqrt{6}}$$

$$|2 - \lambda| = 1$$

$$\lambda = 1, 3$$

$$x^2 + y^2 + 2gx + 2fy = 0$$

$$10 + 29 + 6f = 0$$

$$g + 3f = -5$$

$$3g + f = -5$$

$$9g + 3f = -15$$

$$-8g = 10$$

$$g = -\frac{5}{4}f = -\frac{5}{4}$$

$$r = \sqrt{\frac{25}{16} + \frac{25}{16}}$$

$$= \frac{5\sqrt{2}}{4}$$

**Question:** Probability of event A is 0.7 and event B is 0.4,  $P(A \cap BC) = 0.5$ , then the value  $P(B|A \cup BC)$  is equal to

**Options:**

(a)  $\frac{1}{2}$

(b)  $\frac{1}{3}$

(c)  $\frac{1}{4}$

(d)  $\frac{3}{4}$

**Answer:** (c)

**Question:**  $\int_{-1}^{3/2} |\pi^2 x \sin(\pi x)| dx$

**Options:**

(a)  $4\pi + 1$

(b)  $3\pi + 1$

(c)  $5\pi + 1$

(d)  $6\pi + 1$

**Answer: (b)**

**Question:** The product of last 2 digits of  $(1919)^{1919}$  is

**Options:**

(a) 56

(b) 63

(c) 45

(d) 54

**Answer: (b)**

**Question:** There are 12 points in a plane in which 5 are collinear such that no three of them are in straight line, then the number of triangles that can be formed from any 3 vertices from 12 points

**Options:**

(a) 220

(b) 210

(c) 230

(d) 240

**Answer: (b)**

$$A = \begin{bmatrix} 2 & 2+p & 2+p+q \\ 4 & 6+2p & 8+3p+2q \\ 6 & 12+3p & 20+6p+3q \end{bmatrix},$$

**Question:** If  $\det(\text{adj}(\text{adj}(3A))) = 2^m \cdot 3^n$ , then the value of  $m + n$  is equal to

**Options:**

(a) 20

(b) 24

(c) 36

(d) 18

**Answer: (b)**

**Question:** If  $f(x) = x - 1$  and  $g(x) = ex$  and  $\frac{dy}{dx} = \left( e^{-2\sqrt{x}} g(f(f(x))) - \frac{y}{\sqrt{x}} \right)$ , where  $y(0) = 0$ . Then  $y(1)$  equals to

**Options:**

(a)  $\frac{2e-1}{e^4}$

(b)  $\frac{e-1}{e^4}$

(c)  $\frac{e^3-1}{e^4}$

(d)  $\frac{e^2-1}{e^4}$

**Answer: (b)**

**Question:** Value of  $\cot^{-1}\left(\frac{\sqrt{1+\tan^2 2}+1}{\tan 2}\right) - \cot^{-1}\left(\frac{\sqrt{1+\tan^2 2}-1}{\tan 2}\right)$  is

**Options:**

(a)  $\frac{\pi}{2} + \frac{5}{2}$

(b)  $\frac{\pi}{2} - \frac{3}{2}$

(c)  $2 - \frac{\pi}{2}$

(d)  $3 + \frac{\pi}{2}$

**Answer:** (c)

**Question:** The sum of squares of roots of  $|x-2|^2 - |x-2| - 2 = 0$  and  $x^2 - 2|x-3| - 5 = 0$  equals to

**Answer:** (42)

